ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT

College of Maritime Transport & Technology — Nautical Department

MODEL ANSWER

Introduction to Navigation (BS131) — 7th Week Exam

Date: 06 April 2025 | Duration: 1 Hour | Total: 20 Marks

Detailed worked solutions with rules, intermediate steps, and final answers

Question (1) — 7 Marks | CLO (1), (2) · PLO (a), (d)

A) Complete the following sentences

1-a. Parallel of latitude is a small circle, the plane of which is parallel to the plane of …….

A parallel of latitude is formed by the intersection of the Earth with a plane parallel to the equatorial plane. Therefore every parallel lies parallel to the equator.

Answer: the Equator

1-b. Rhumb line is the line on the earth's surface which cuts all ……. at the same angle.

By definition, a rhumb line (loxodrome) maintains a constant true course, so it intersects every meridian at the same angle.

Answer: Meridians (meridians of longitude)

1-c. Meridians are semi great circles joining ……. and are perpendicular to the equator.

Each meridian is a semi-great circle running from the North Pole to the South Pole and cutting the equator at 90°.

Answer: the North and South Poles (the poles)

B) Position calculations

B-1. Calculate D.Lat and D.Long between Pos. A and Pos. B

Pos. A = 28° 30.8′ S , 057° 28.6′ W

Pos. B = 29° 42.6′ N , 069° 33.4′ W

Difference of Latitude (D.Lat)

Sign convention used: North (+), South (–).

```
Lat A = –28° 30.8′    Lat B = +29° 42.6′
D.Lat = Lat B – Lat A = 29° 42.6′ – (–28° 30.8′) = 29° 42.6′ + 28° 30.8′
D.Lat = 58° 13.4′ North  (vessel changes latitude northward from A to B)
```

Difference of Longitude (D.Long)

Both longitudes are West; Pos. B is further west than Pos. A.

```
D.Long = 069° 33.4′ W – 057° 28.6′ W = 012° 04.8′ West
```

Final Answer (B-1): D.Lat = 58° 13.4′ N | D.Long = 12° 04.8′ W

B-2. Find the coordinates of the final position

Pos. A = 33° 24.8′ N , 172° 15.6′ E

D.Lat = 22° 56.7′ S D.Long = 011° 57.8′ E

Final Latitude

```
Final Lat = 33° 24.8′ N − 22° 56.7′ = 10° 28.1′ N
```

Final Longitude

```
Final Long (raw) = 172° 15.6′ E + 011° 57.8′ E = 184° 13.4′ E
```

Longitude cannot exceed 180°. Convert the excess over 180° to West longitude:

```
184° 13.4′ E = (360° − 184° 13.4′) W = 175° 46.6′ W
```

Alternatively: excess east of 180° is 4° 13.4′, so West = 180° – 4° 13.4′ = 175° 46.6′ W.

Final Answer (B-2): 10° 28.1′ N , 175° 46.6′ W

Question (2) — 6 Marks | CLO (3), (4), (6) · PLO (a), (d)

A) Extract the following values from the figure

From the printed diagram: Magnetic North (M.N) lies **6° West** of True North (T.N); Compass North (C.N) lies **2° East** of T.N; the heading arc from C.N is **085°**; the lighthouse bearing arc from T.N is **122°**.

1) Variation

Variation is the angle between True North and Magnetic North.

```
M.N is 6° to the West of T.N ⇒ Variation = 6° W
```

Variation = 6° W

2) Total Compass Error

Total Compass Error (C.E.) is the angle between True North and Compass North.

```
C.N is 2° to the East of T.N ⇒ Total C.E. = 2° E
```

Check rule: Error East, Compass Least — compass courses will be less than true courses.

Total Compass Error = 2° E

3) Compass Course

The figure labels the course arc from Compass North to the ship's heading as 085°.

Compass Course = 085° C

4) True Course

Apply total compass error (Error East → add to compass to obtain true):

```
True Course = Compass Course + C.E. (E) = 085° + 2° = 087° T
```

Cross-check via Magnetic Course:

```
C.E. = Var + Dev ⇒ +2° = (−6°) + Dev ⇒ Dev = 8° E
Mag. Course = Comp. Course + Dev (E) = 085° + 8° = 093° M
True Course = Mag. Course − Var (W) = 093° − 6° = 087° T ✓
```

True Course = 087° T

5) True Bearing

The figure labels the bearing arc from True North to the lighthouse line as 122°.

True Bearing = 122° T

6) Relative Bearing

Relative bearing = True Bearing – True Course (starboard / Green if positive).

```
Relative Bearing = 122° − 087° = 035° Green (Starboard)
```

Relative Bearing = 035° Green (to Starboard)

Summary of extracted / calculated values

Quantity	Value
Variation	6° W
Total Compass Error	2° E
Compass Course	085° C
True Course	087° T
True Bearing	122° T
Relative Bearing	035° Green

B) Insert the appropriate term (Gnomonic Projection — North Polar Aspect)

On a gnomonic chart a straight line is a **great-circle track**. The Seattle–London track shows an initial course near Seattle of about **029°** and a final course near London of about **129°**. The three marked points on the track are identified as follows:

1 → point near departure / start of the GC track → **Initial Course**

2 → highest-latitude point of the great circle (nearest the pole) → **Vertex**

3 → point near arrival / end of the GC track → **Final Course**

1. Initial Course (Initial Great Circle Course) 2. Vertex 3. Final Course (Final Great Circle Course)

Note: Intermediate open circles at successive meridians are also the transfer waypoints used when converting the GC track to rhumb-line legs on a Mercator chart.

Question (3) — 7 Marks | CLO (3), (6) · PLO (a), (d)

A) Fill in the missed values of the following table

Rule used (Compass ↔ Magnetic): Mag. Course = Comp. Course ± Deviation, where **East deviation is added** to compass to obtain magnetic, and **West deviation is subtracted** (equivalently: Deviation East ⇒ Compass Least).

Row 1: C.Course = 155°, Dev. = 3° W, Mag.Course = ?

```
Mag. Course = 155° – 3° (W) = 152° M
```

Row 2: Dev. = 5° E, Mag.Course = 273°, C.Course = ?

```
Comp. Course = 273° – 5° (E) = 268° C
```

Check: 268° + 5° E = 273° M ✓

	C.Course	Dev.	Mag.Course
1	155°	3° W	**152° M**
2	**268° C**	5° E	273°

Row 1 Mag.Course = 152° M | Row 2 C.Course = 268° C

B) Relative / True bearings

B-1. T.Co = 023°, Relative bearing of lighthouse = 045° **Red**. Find T.Bg.

*Convention: **Red = Port (left)** → subtract relative bearing from the ship's true course. **Green = Starboard (right)** → add.*

```
T.Bg = T.Co – Rel.Bg (Red) = 023° – 045° = –022°
Normalize to the 000°–360° range: –022° + 360° = 338°
```

True Bearing of the lighthouse = 338° T

B-2. T.Co = 111°, True bearing of lighthouse = 140°. Find Relative bearing.

```
Relative Bearing = T.Bg – T.Co = 140° – 111° = 029°
```

Positive result ⇒ object is on the starboard side ⇒ Green.

Relative Bearing = 029° Green (Starboard)

C) Compass & Gyro errors

Given: True Course = 333° T | Compass Course = 322° C | Gyro Course = 335° | Deviation = 5° E

C-1. Total Compass Error

```
C.E. = True Course – Compass Course = 333° – 322° = 11°
```

Compass is less than True ⇒ Error is Easterly (*Error East, Compass Least*).

Total Compass Error = 11° E

C-2. Variation

Using signed values (East +, West –): C.E. = Variation + Deviation

```
+11° = Variation + (+5°) ⇒ Variation = +6° = 6° E
```

Check:

```
Mag. Course = Comp. + Dev (E) = 322° + 5° = 327° M
True Course = Mag. + Var (E) = 327° + 6° = 333° T ✓
```

Variation = 6° E

C-3. Gyro Error

```
Gyro Error = Gyro Course – True Course = 335° – 333° = 2°
```

Gyro reading is greater than true ⇒ Gyro is **High** by 2°.

Gyro Error = 2° High (2° H)

C-4. Compass bearing & Gyro bearing if True Bearing = 359° T

Apply the same errors found above (errors are constant for the given heading situation):

```
Compass Bearing = True Bearing – C.E. (E) = 359° – 11° = 348° C
Gyro Bearing = True Bearing + Gyro Error (High) = 359° + 2° = 361° ≡ 001° G
```

Compass Bearing = 348° C | Gyro Bearing = 001° G

Quick Reference — Final Answers Only

Item	Final Answer
Q1-A a / b / c	Equator / Meridians / North & South Poles
Q1-B1 D.Lat / D.Long	58° 13.4′ N / 12° 04.8′ W
Q1-B2 Final Position	10° 28.1′ N , 175° 46.6′ W
Q2-A (Var / CE / C.Co / T.Co / T.Bg / Rel)	6°W / 2°E / 085°C / 087°T / 122°T / 035°G

Q2-B Terms 1 / 2 / 3	Initial Course / Vertex / Final Course
Q3-A Mag / Comp	152° M / 268° C
Q3-B1 T.Bg	338° T
Q3-B2 Rel.Bg	029° Green
Q3-C CE / Var / GE	11°E / 6°E / 2° High
Q3-C4 Comp.Bg / Gyro.Bg	348° C / 001° G

End of Model Answer — Introduction to Navigation (BS131) · 7th Week Exam · 20 Marks
Rules applied: Cadet (Compass Add East True) · Error East/Compass Least · Red = Port / Green = Starboard · Gyro High/Low